

Limiti Notevoli

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\tan(x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \log_a e$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

ESERCIZIO n°1

$$\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} = \frac{0}{0} \text{ INDETERMINATO}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} = \frac{\overset{1}{\sin(x)} \cdot \overset{1}{\sin(x)} \cdot \overset{0}{2x}}{x \cdot x} = 0$$

ESERCIZIO 2

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \frac{3x}{x} = 3$$

$$\begin{aligned} \sin 3x &\sim 3x \\ x &\sim x \end{aligned}$$

ESERCIZIO 3

$$0 \quad 1 - \cos x \quad 1 - 1 = 0$$

$$\lim_{x \rightarrow 0} \frac{x^2}{x^2} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1 - \cos x}{x} \cdot \frac{1 + \cos x}{1 + \cos x} = \frac{1 - \cos^2 x}{x^2} \cdot \frac{1}{1 + \cos x}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \cdot \frac{1}{1 + \cos x} = 1^2 \cdot \frac{1}{1 + 1} = \frac{1}{2}$$

$\frac{\sin x}{x} = 1$

ESERCIZIO 10

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \frac{1 - 1}{0} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \frac{1 - \cos x}{x} \cdot \frac{1}{x} \cdot \frac{x}{2} = 0$$

ES. 5

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 4x)}{x} = \frac{0}{0}$$

1° METODO

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 4x)}{x} = 4 \frac{\ln(1 + x)}{x} = 4 \cdot 1 = 4$$

2° METODO

$$4 \ln(1 + \frac{1}{4x})$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1 \quad \Rightarrow \quad 1 \cdot 1 = 1$$

ES 6

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^2} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^2} = \frac{\sin x - \frac{\sin x}{\cos x} \cdot \frac{1}{x}}{x - x} = 1 - \frac{1}{\cos x} = 0$$

ES 7

$$\lim_{x \rightarrow 0} \left(\frac{1}{\tan x} - \frac{1}{\sin x} \right) = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{\frac{\sin x}{\cos x}} - \frac{1}{\sin x} \right) \Rightarrow \frac{\sin x - \tan x}{\sin x - \tan x} \Rightarrow \frac{\sin x - \frac{\sin x}{\cos x}}{\sin x - \frac{\sin x}{\cos x}} \cdot \frac{x^3}{x^3} \cdot \cos x = 0$$

ES 8

$$\lim_{x \rightarrow -\infty} x(3^{\frac{1}{x}} - 1) = \frac{0}{0}$$

IMPONERO

$$\frac{1}{x} = t$$

$$\lim_{x \rightarrow 0^-} \frac{3t - 1}{t} = \log 3$$

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{x} = \log_e 2$$

BS. 9

$$\lim_{x \rightarrow 2} \frac{x-2}{\log(\frac{x}{2})} = \frac{0}{0}$$

$$\lim_{t \rightarrow 0} \frac{t}{\log(\frac{t+2}{2})} \Rightarrow 2 \cdot \frac{\frac{t}{2}}{\log(1+\frac{t}{2})} = 2$$

$$x-2=t$$

BS. 10

$$\lim_{x \rightarrow +\infty} \frac{e^{3x} + 1}{e^{2x} + 2} = \frac{0}{0}$$

$$e^x = t$$

$$\lim_{x \rightarrow +\infty} \frac{e^{3x} + 1}{e^{2x} + 2} = \frac{t^3 + 1}{t^2 + 2} = +\infty$$

$$\lim_{x \rightarrow -\infty} \frac{e^{3x} + 1}{e^{2x} + 2} = \frac{\cancel{t^3} + 1}{\cancel{t^2} + 2} = \frac{1}{2}$$

BS. 11

$$\lim_{x \rightarrow 1} \frac{\sqrt{x} - x}{x - 1} = \frac{\sqrt{1} - 1}{1 - 1} = \frac{0}{0} \quad \text{IND}$$